
Consider $f(x) = 3\cos(2x)$.

Evaluate:

$$\begin{aligned}\lim_{x\to\pi/2} f(x) &= 3\lim_{x\to\pi/2}\cos(2x)\\ &= 3\cos\left(\cancel{2}\cdot\frac{\pi}{\cancel{2}}\right)\\ &= -3\end{aligned}$$

Evaluate:

$$\lim_{x\to-\inf} f(x) = 3\lim_{x\to\inf}\cos(2x)$$

Since the limit at infinity of cosines does not exist, this is DNE.

What is the average rate of change of $f(x)$ over $x \in \left[\frac{\pi}{4}, \frac{\pi}{4}+h\right]$ where $h > 0$?

$$\frac{\Delta f}{\Delta x} = ???$$

Evaluate:

$$\begin{aligned}\sin\left(-\frac{5\pi}{4}\right) &= \sin\left(\frac{3\pi}{4}\right)\\ &= \frac{\sqrt{2}}{2}\end{aligned}$$

Evaluate:

$$\begin{aligned}\sec(-240°) &= \sec(120°)\\ &= \cos^{-1}(120°)\\ &= \left(-\frac{1}{2}\right)^{-1}\\ &= -2\end{aligned}$$

Find the slope of the tangent to $f(x) = \ln(2x-1)$ at $x = 3$.

$$\begin{aligned}f_1(x) &= 2x-1\\ f_2(f_1) &= \ln f_1\\ f_{1'}(x)p &= 2\\ f_{2'}(f_1) &= \frac{1}{f_1}\\ f'(x) &= \frac{1}{2x-1}\cdot 2\\ &= \frac{2}{2x-1}\end{aligned}$$

Now we evaluate this at $x = 3$ and get the slope.

$$m = \frac{2}{2 \cdot 3 - 1} = \frac{2}{5}$$

Differentiate $g(x) = \ln(7x - 5)$.

$$\begin{aligned} f_1(x) &= 7x - 5 \\ f_2(x) &= \ln f_1 \\ f_{1'}(x) &= 7 \\ f_{2'}(x) &= \frac{1}{f_1} \\ g'(x) &= \frac{1}{7x-5} \cdot 7 \\ &= \frac{7}{7x-5} \end{aligned}$$

Differentiate $g(x) = \ln x^2$.

We have $f_1 = x^2$ and $f_2 = \ln f_1$.

$$\begin{aligned} f_{1'} &= 2x \\ f_{2'} &= \frac{1}{f_1} g'(x) = f_{2'} \cdot f_{1'} \\ &= \frac{2x}{x^2} \\ &= \frac{2}{x} \end{aligned}$$
